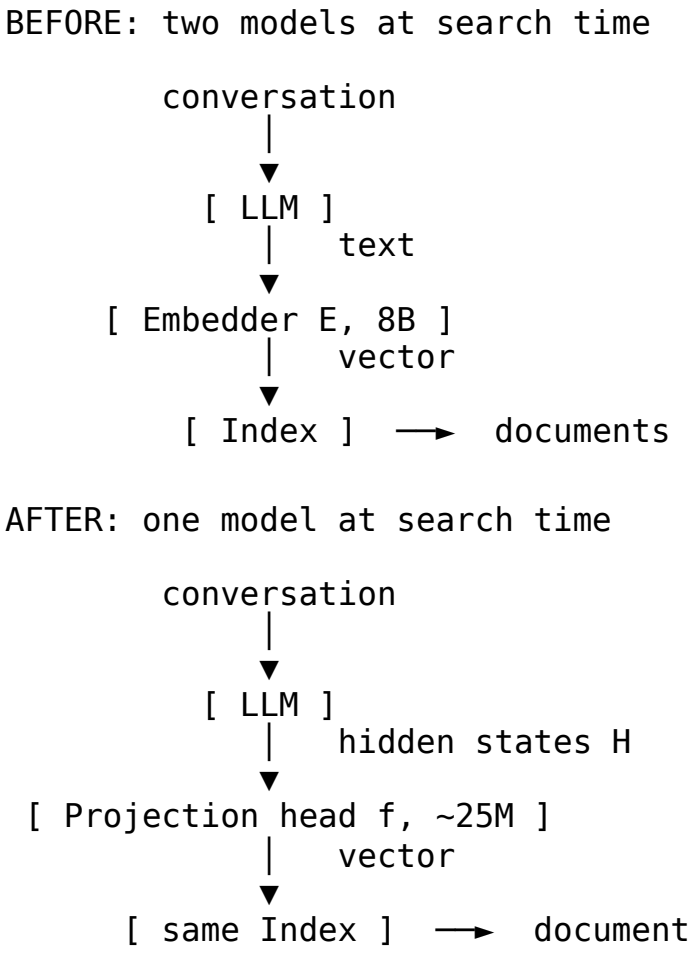
10 · The Solution at a Glance

One sentence

Attach a small network, the **projection head**, to the LLM. It turns the hidden states the LLM produced while writing the query into a vector that lives in the **same space** as the existing embedding model's vectors, so the existing document index can be searched without running the embedder.

Before and after



The index doesn't change. Only the thing that produces the query

vector changes.

The six solution steps

Step	File	Idea	Fixes
1	11	**Core insight:** the LLM already holds what the search needs.	Frames all four problems
2	12	**Extract hidden states** during normal generation, at no extra compute.	Problem 2
3	13	**Projection head:** resize → mix → average → resize → normalize.	Problems 1, 2, 3
4	14	**Learn from a teacher:** train the head to copy the existing embedder, so the old index still works.	Problem 1 (keeps the index)
5	15	**Three losses:** alignment, contrastive, rank distillation, each catching a different mistake.	Makes the copy good enough to use
6	16	**Training recipe and deployment:** cache traces, train slowly, then drop the teacher.	Problems 1, 2 at inference

The names to remember

Name	Symbol	Role
LLM (agent)	—	Qwen3-8B. Writes queries, produces hidden states. **Frozen**, never trained here.
Teacher	E	Qwen3-Embedding-8B. Built the index. Gives the "correct" query vectors during training. **Frozen.** Not needed after training.
Student	f	The projection head. The **only** part that is trained.

Name	Symbol	Role
Hidden states	$\mathbf{H}$	Input to the student.
Teacher vector	$\mathbf{y} = E(q)$	The target the student is trained to match.

The math, in one place

The worked example follows this whole pipeline with tiny numbers. Start with `math/00_the_cast.md` and `math/01_math_toolkit.md` if you haven't already.